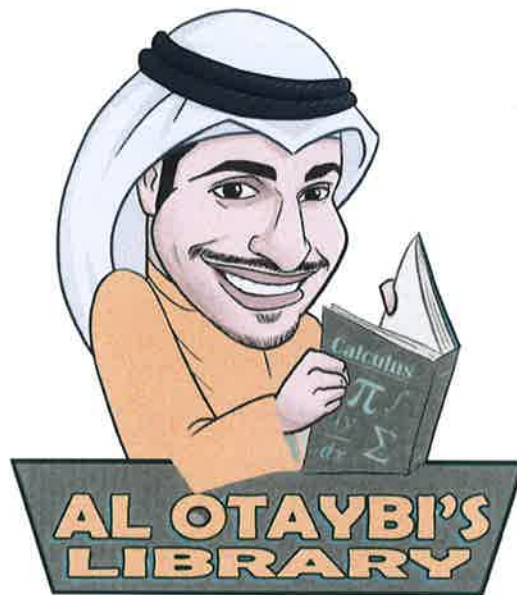


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.2

11.2 Series.

(9)



11.2 Series



Limit of sequences given by recurrence relation

11.1 Sequences

$$\{a_n\} = \{a_1, a_2, a_3, a_4, \dots, a_n\}$$

مجموعة من الحدود تربط بينهما علاقة معينة
(بينهما فواصل)

Example

$$\left\{\frac{1}{n}\right\} = \left\{\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$$

11.2 Series

$$\sum_{n=i}^{\infty} a_n = a_1 + a_2 + a_3 + a_4 + \dots + a_n$$

Sequences ————— مجموع حدود الـ
(بينهما جمع)

Example

$$\sum_{n=1}^{\infty} \frac{1}{n} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \dots$$



Telescoping



Limit of sequences given by recurrence relation

ومن نفس النوع
الفرق بينهم عدد

(حد - حد)

انتبه !!

توزيع
المقام

Partial
fraction

خواص
 $\ln$

Examples

$$\sum \left[\sin \frac{1}{n} - \sin \frac{1}{n+1} \right]$$

$$\sum [\tan^{-1}(n) - \tan^{-1}(n+2)]$$

$$\sum \left[\frac{1}{e^n} - \frac{1}{e^{n+1}} \right]$$

$$\sum \ln \left(\frac{n}{n+1} \right)$$

$$\sum \frac{2}{n^2 + 4n + 3}$$

$$\sum \frac{\sqrt{n+1} - \sqrt{n-1}}{\sqrt{n+1} \sqrt{n-1}}$$



Telescoping

خطوات حل مسائل



(1) كتابة السيريس على شكل التيليسكوبينق

$$a_n = \dots$$

(2) كتابة

$$S_n = \dots$$

(3) استنتاج مع مراعاة بداية السيريس

$$S_n$$

(4) ايجاد اللمت للـ

$$\lim_{x \rightarrow \infty} S_n$$

$\pm\infty$ or DNE

رقم

Divergent

Convergent

No sum

Sum = عدد

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=1}^{\infty} \left[\sin \frac{1}{n} - \sin \frac{1}{n+1} \right]$$

$$a_n = \sin \frac{1}{n} - \sin \frac{1}{n+1}$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \left[\sin \frac{1}{1} - \sin \frac{1}{2} \right] + \left[\sin \frac{1}{2} - \sin \frac{1}{3} \right] + \left[\sin \frac{1}{3} - \sin \frac{1}{4} \right] + \dots + \left[\sin \frac{1}{n-2} - \sin \frac{1}{n-1} \right] + \left[\sin \frac{1}{n-1} - \sin \frac{1}{n} \right] + \left[\sin \frac{1}{n} - \sin \frac{1}{n+1} \right]$$

$$\Rightarrow \therefore S_n = \sin 1 - \sin \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\sin 1 - \sin \frac{1}{n+1} \right]$$

$$= \sin 1 - \sin \frac{1}{\infty}$$

$$= \sin 1 - \sin 0 = \sin 1$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n \text{ is convergent (Telescoping)}$$

$$\text{and the Sum} = \sin 1$$

Note

(حد - حد)
ومن نفس النوع

Telescoping



انتبه من البداية



Text book question



Determine whether the series is convergent or divergent.

If convergent, find its sum.

$$\sum_{n=4}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right) \quad \text{with a red arrow pointing to } a_n$$

Let $a_n = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$

$$S_n = a_4 + a_5 + \dots + a_{n-1} + a_n$$

$$= \left[\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{5}} \right] + \left[\frac{1}{\sqrt{5}} - \frac{1}{\sqrt{6}} \right] + \dots + \left[\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}} \right] + \left[\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right]$$

$$\therefore S_n = \frac{1}{2} - \frac{1}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{1}{2} - \frac{1}{\sqrt{n+1}} \right] = \frac{1}{2} - 0 = \frac{1}{2}$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Telescoping)

and the sum = $\frac{1}{2}$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=0}^{\infty} [\tan^{-1}(n) - \tan^{-1}(n+2)]$$

انتبه من البداية

$$a_n = \tan^{-1}(n) - \tan^{-1}(n+2)$$

$$S_n = a_0 + a_1 + a_2 + \dots + a_{n-2} + a_{n-1} + a_n$$

Note
(حد - حد)
ومن نفس النوع

$$= [\cancel{\tan^{-1}(0)} - \cancel{\tan^{-1}(2)}] + [\cancel{\tan^{-1}(1)} - \cancel{\tan^{-1}(3)}] + [\cancel{\tan^{-1}(2)} - \cancel{\tan^{-1}(4)}] + \dots + [\cancel{\tan^{-1}(n-2)} - \cancel{\tan^{-1}(n-2+2)}] + [\cancel{\tan^{-1}(n-1)} - \cancel{\tan^{-1}(n+2)}] + [\cancel{\tan^{-1}(n)} - \cancel{\tan^{-1}(n+2)}]$$

$$\Rightarrow S_n = \tan^{-1}(0) + \tan^{-1}(1) - \tan^{-1}(n+1) - \tan^{-1}(n+2)$$

$$= \frac{\pi}{4} - \tan^{-1}(n+1) - \tan^{-1}(n+2)$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{\pi}{4} - \tan^{-1}(n+1) - \tan^{-1}(n+2) \right]$$

$$= \frac{\pi}{4} - \frac{\pi}{2} - \frac{\pi}{2} = \frac{\pi}{4} - \frac{2\pi}{4} - \frac{2\pi}{4} = -\frac{3\pi}{4}$$

$\Rightarrow \therefore \sum_{n=0}^{\infty} a_n$ is convergent (Telescoping)

and the Sum = $-\frac{3\pi}{4}$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=1}^{\infty} \left[e^{\frac{1}{n}} - e^{\frac{1}{n+1}} \right]$$

Let $a_n = e^{\frac{1}{n}} - e^{\frac{1}{n+1}}$

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \left[e^1 - e^{\frac{1}{2}} \right] + \left[e^{\frac{1}{2}} - e^{\frac{1}{3}} \right] + \left[e^{\frac{1}{3}} - e^{\frac{1}{4}} \right] + \dots + \left[e^{\frac{1}{n-2}} - e^{\frac{1}{n-1}} \right] + \left[e^{\frac{1}{n-1}} - e^{\frac{1}{n}} \right] + \left[e^{\frac{1}{n}} - e^{\frac{1}{n+1}} \right]$$

$$\Rightarrow S_n = e^1 - e^{\frac{1}{n+1}}$$

$$\lim_{n \rightarrow \infty} S_n$$

$$\lim_{n \rightarrow \infty} \left[e^1 - e^{\frac{1}{n+1}} \right] = e^1 - e^{\frac{1}{\infty}} = e^1 - e^0 = e - 1$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \left[e^{\frac{1}{n}} - e^{\frac{1}{n+1}} \right] \text{ is convergent}$$

(Telescoping)

$$\text{Sum} = e - 1$$

Determine whether the series is convergent or divergent.
If converges, find its sum.

$$\sum_{n=1}^{\infty} \frac{6}{(3n+2)(3n+5)} \rightarrow a_n$$

Let $a_n = \frac{6}{(3n+2)(3n+5)}$

$$\frac{6}{(3x+2)(3x+5)} = \frac{A}{3x+2} + \frac{B}{3x+5} \quad * (3x+2)(3x+5)$$

$$6 = A(3x+5) + B(3x+2)$$

$$x = -\frac{2}{3} \rightarrow 6 = A(\cancel{3(-\frac{2}{3})} + 5) + 0 \rightarrow 6 = 3A \rightarrow \boxed{A=2}$$

$$x = -\frac{5}{3} \rightarrow 6 = 0 + B(\cancel{3(-\frac{5}{3})} + 2) \rightarrow 6 = -3B \rightarrow \boxed{B=-2}$$

$$\therefore a_n = \frac{2}{3n+2} - \frac{2}{3n+5}$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \left[\frac{2}{5} - \frac{2}{8} \right] + \left[\frac{2}{8} - \frac{2}{11} \right] + \left[\frac{2}{11} - \frac{2}{14} \right] + \dots +$$

$$\left[\frac{2}{3(n-2)+2} - \frac{2}{3(n-2)+5} \right] + \left[\frac{2}{3(n-1)+2} - \frac{2}{3(n-1)+5} \right] + \left[\frac{2}{3n+2} - \frac{2}{3n+5} \right]$$

$$\therefore S_n = \frac{2}{5} - \frac{2}{3n+5}$$

$$\lim_{n \rightarrow \infty} S_n$$

$$\lim_{n \rightarrow \infty} \left[\frac{2}{5} - \frac{2}{3n+5} \right] = \frac{2}{5} - \frac{2}{\infty} = \frac{2}{5} - 0 = \frac{2}{5}$$

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is convergent (Telescoping)}$$

$$\text{and the Sum} = \frac{2}{5}$$



Text book question



Determine whether the series is convergent or divergent.

If convergent, find its sum.

$$\sum_{n=2}^{\infty} \frac{2}{n^2 - 1} \quad \rightarrow a_n$$

$$a_n = \frac{2}{n^2 - 1}$$

$$\frac{2}{x^2 - 1}$$

$$\frac{2}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$* (x-1)(x+1)$$

$$2 = A(x+1) + B(x-1)$$

$$\underline{x=1} \quad 2 = A(2) + 0 \rightarrow A = 1$$

$$\underline{x=-1} \quad 2 = 0 + B(-2) \rightarrow B = -1$$

$$\therefore a_n = \frac{1}{n-1} - \frac{1}{n+1}$$

$$S_n = a_2 + a_3 + a_4 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \left[\frac{1}{1} - \frac{1}{3} \right] + \left[\frac{1}{2} - \frac{1}{4} \right] + \left[\frac{1}{3} - \frac{1}{5} \right] + \dots +$$

$$\left[\frac{1}{n-2} - \frac{1}{n-1} \right] + \left[\frac{1}{n-1} - \frac{1}{n} \right] + \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$\therefore S_n = 1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = 1 + \frac{1}{2} - 0 - 0 = \frac{3}{2}$$

$\therefore \sum_{n=2}^{\infty} a_n$ is convergent (telescoping) and the sum = $\frac{3}{2}$

Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

Spring 2019

Let $a_n = \frac{1}{n(n+2)}$

$$f(x) = \frac{1}{x(x+2)}$$

$$\frac{1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2}$$

* $x(x+2)$

$$1 = A(x+2) + Bx$$

$x=0$ $1 = A(2) + 0 \rightarrow A = \frac{1}{2}$

$x=-2$ $1 = 0 + B(-2) \rightarrow B = -\frac{1}{2}$

$$\Rightarrow \therefore a_n = \frac{\frac{1}{2}}{n} - \frac{\frac{1}{2}}{n+2} = \frac{1}{2} \left[\frac{1}{n} - \frac{1}{n+2} \right]$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \frac{1}{2} \left[\left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \dots + \left(\frac{1}{n-2} - \frac{1}{n-2+2} \right) + \left(\frac{1}{n-1} - \frac{1}{n-1+2} \right) + \left(\frac{1}{n} - \frac{1}{n+2} \right) \right]$$

$$= \frac{1}{2} \left[1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{2} \left[1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right] \right] = \frac{1}{2} \left[1 + \frac{1}{2} - 0 - 0 \right] = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent and the sum = $3/4$ (Telescoping)

Use definition to find the sum of the series

$$\sum_{n=1}^{\infty} \frac{(n+1)\ln n - n\ln(n+1)}{n(n+1)}$$

$$\sum_{n=1}^{\infty} \left[\frac{\cancel{(n+1)} \ln n}{\cancel{n} \cancel{(n+1)}} - \frac{\cancel{n} \ln(n+1)}{\cancel{n} \cancel{(n+1)}} \right]$$

$$\sum_{n=1}^{\infty} \left[\frac{\ln n}{n} - \frac{\ln(n+1)}{n+1} \right]$$

$$a_n = \frac{\ln n}{n} - \frac{\ln(n+1)}{n+1}$$

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n \\ &= \left[\frac{\ln 1}{1} - \frac{\ln 2}{2} \right] + \left[\frac{\ln 2}{2} - \frac{\ln 3}{3} \right] + \left[\frac{\ln 3}{3} - \frac{\ln 4}{4} \right] + \dots + \\ &\quad \left[\frac{\ln(n-2)}{n-2} - \frac{\ln(n-1)}{n-1} \right] + \left[\frac{\ln(n-1)}{n-1} - \frac{\ln n}{n} \right] + \left[\frac{\ln n}{n} - \frac{\ln(n+1)}{n+1} \right] \end{aligned}$$

$$\Rightarrow S_n = \frac{\ln 1}{1} - \frac{\ln(n+1)}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[0 - \frac{\ln(n+1)}{n+1} \right] \quad \frac{0}{\infty}$$

Let $f(x) = -\frac{\ln(x+1)}{x+1}$

$$\lim_{x \rightarrow \infty} -\frac{\ln(x+1)}{x+1} \quad \text{L'H} \quad \lim_{x \rightarrow \infty} -\frac{\frac{1}{x+1}}{\frac{1}{x+1}} = 0$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n \text{ is convergent (Telescoping)} \Rightarrow \text{Sum} = 0$$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=2}^{\infty} \frac{\sqrt{n+1} - \sqrt{n-1}}{\sqrt{n+1}\sqrt{n-1}}$$

$$\sum_{n=2}^{\infty} \left[\frac{\sqrt{n+1}}{\sqrt{n+1}\sqrt{n-1}} - \frac{\sqrt{n-1}}{\sqrt{n+1}\sqrt{n-1}} \right]$$

$$\sum_{n=2}^{\infty} \left[\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n+1}} \right]$$

Let $a_n = \frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n+1}}$

$$S_n = a_2 + a_3 + a_4 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$= \left[\frac{1}{1} - \frac{1}{\sqrt{3}} \right] + \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{4}} \right] + \left[\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{5}} \right]$$

$$+ \dots + \left[\frac{1}{\sqrt{n-3}} - \frac{1}{\sqrt{n-1}} \right] + \left[\frac{1}{\sqrt{n-2}} - \frac{1}{\sqrt{n}} \right] + \left[\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n+1}} \right]$$

$$\Rightarrow S_n = 1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} \left[1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right] = 1 + \frac{1}{\sqrt{2}}$$

$\Rightarrow \therefore \sum_{n=2}^{\infty} a_n$ is convergent (Telescoping)

$$\rightarrow \text{Sum} = 1 + \frac{1}{\sqrt{2}}$$

Determine whether the series is convergent or divergent . if it converges find its sum

$$\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$$

$$\sum_{n=1}^{\infty} \ln\left(\frac{n}{n} + \frac{1}{n}\right) = \sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$$

$$\sum_{n=1}^{\infty} [\ln(n+1) - \ln(n)]$$

$$\sum_{n=1}^{\infty} -[\ln(n) - \ln(n+1)]$$

$$a_n = -[\ln(n) - \ln(n+1)]$$

$$S_n = -[a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n]$$

$$= -[(\ln 1 - \ln 2) + (\ln 2 - \ln 3) + (\ln 3 - \ln 4) + \dots + (\ln(n-2) - \ln(n-1)) + (\ln(n-1) - \ln(n)) + (\ln(n) - \ln(n+1))]$$

$$S_n = -[\ln 1 - \ln(n+1)]$$

$$S_n = \ln(n+1)$$

$$\Rightarrow \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \ln(n+1) = \ln \infty = \infty$$

$\therefore \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$ is Divergent (Telescoping)
and No Sum

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$$

$$\begin{aligned} a_n &= \ln\left(\frac{n^2}{n^2} - \frac{1}{n^2}\right) = \ln\left(\frac{n^2-1}{n^2}\right) \\ &= \ln\left(\frac{(n-1)(n+1)}{n \cdot n}\right) = \ln\left(\frac{n-1}{n}\right) + \ln\left(\frac{n+1}{n}\right) \\ &= \ln\left(\frac{n-1}{n}\right) + \ln\left(\frac{n}{n+1}\right)^{-1} \end{aligned}$$

$$\Rightarrow a_n = \ln\left(\frac{n-1}{n}\right) - \ln\left(\frac{n}{n+1}\right)$$

$$S_n = a_2 + a_3 + a_4 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$\begin{aligned} &\left[\ln\left(\frac{1}{2}\right) - \ln\left(\frac{2}{3}\right)\right] + \left[\ln\left(\frac{2}{3}\right) - \ln\left(\frac{3}{4}\right)\right] + \left[\ln\left(\frac{3}{4}\right) - \ln\left(\frac{4}{5}\right)\right] + \dots + \\ &\left[\ln\left(\frac{n-2}{n-1}\right) - \ln\left(\frac{n-1}{n}\right)\right] + \left[\ln\left(\frac{n-1}{n}\right) - \ln\left(\frac{n}{n+1}\right)\right] + \left[\ln\left(\frac{n}{n+1}\right) - \ln\left(\frac{n+1}{n+2}\right)\right] \end{aligned}$$

$$\Rightarrow S_n = \ln\left(\frac{1}{2}\right) - \ln\left(\frac{n}{n+1}\right)$$

$$\lim_{n \rightarrow \infty} \left[\ln\left(\frac{1}{2}\right) - \ln\left(\frac{n}{n+1}\right) \right]$$

$$= \lim_{n \rightarrow \infty} \left[\ln\left(\frac{1}{2}\right) - \ln\left(\frac{n}{n(1+\frac{1}{n})}\right) \right]$$

$$= \ln\left(\frac{1}{2}\right) - \ln\left(\frac{1}{1+0}\right) = \ln\frac{1}{2} - \ln 1 = \ln\frac{1}{2}$$

$$\Rightarrow \therefore \sum_{n=2}^{\infty} a_n \text{ is convergent (Telescoping)} \rightarrow \text{Sum} = \ln\frac{1}{2}$$

Determine whether the series is convergent or divergent.

If it convergent find its sum

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+4} + \sqrt{n+3}}$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+4} + \sqrt{n+3}} * \frac{\sqrt{n+4} - \sqrt{n+3}}{\sqrt{n+4} - \sqrt{n+3}}$$

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+4} - \sqrt{n+3}}{n+4 - (n+3)} \Rightarrow n+4 - n - 3 = 1$$

$$\sum_{n=1}^{\infty} (\sqrt{n+4} - \sqrt{n+3}) = \sum_{n=1}^{\infty} -(\sqrt{n+3} - \sqrt{n+4})$$

$$a_n = -(\sqrt{n+3} - \sqrt{n+4})$$

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n \\ &= - \left[(\sqrt{4} - \sqrt{5}) + (\sqrt{5} - \sqrt{6}) + (\sqrt{6} - \sqrt{7}) + \dots + \right. \\ &\quad \left. (\sqrt{n+1} - \sqrt{n+2}) + (\sqrt{n+2} - \sqrt{n+3}) + (\sqrt{n+3} - \sqrt{n+4}) \right] \end{aligned}$$

$$\Rightarrow S_n = -[\sqrt{4} - \sqrt{n+4}]$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} -[2 - \sqrt{n+4}] = -[-\infty] = \infty$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is Divergent (Telescoping)
 $\Rightarrow$ No Sum.

Let $\{b_n\}_{n=1}^{\infty}$ be a convergent sequence with $\lim_{n \rightarrow \infty} b_n = L$

(a) Find the n-th partial sum of the series $\sum_{n=1}^{\infty} (b_n - b_{n+1})$

(b) Show that the series $\sum_{n=1}^{\infty} (b_n - b_{n+1})$ is convergent, and find its sum

$$(a) \sum_{n=1}^{\infty} (b_n - b_{n+1})$$

$$\text{Let } a_n = b_n - b_{n+1}$$

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n \\ &= [b_1 - b_2] + [b_2 - b_3] + [b_3 - b_4] + \dots + \\ &\quad [b_{n-2} - b_{n-1}] + [b_{n-1} - b_n] + [b_n - b_{n+1}] \end{aligned}$$

$$\Rightarrow \therefore S_n = b_1 - b_{n+1}$$

$$\begin{aligned} (b) \quad \lim_{n \rightarrow \infty} S_n &= \lim_{n \rightarrow \infty} (b_1 - b_{n+1}) \\ &= b_1 - L \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} b_n = L$$

$$\therefore \lim_{n \rightarrow \infty} b_{n+1} = L$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} (b_n - b_{n+1})$ is convergent (Telescoping)
and its sum = $b_1 - L$



اذا عطاني S_n و طلب a_n

$$a_n = S_n - S_{n-1}$$



If the n th partial sum of the convergent series

$$\sum_{n=1}^{\infty} a_n \quad \text{is} \quad S_n = 3 - n2^{-n}$$

Find a_n and find the sum of $\sum_{n=1}^{\infty} a_n$

$$a_n = S_n - S_{n-1}$$

$$S_n = 3 - n2^{-n} \longrightarrow S_{n-1} = 3 - (n-1)2^{-(n-1)}$$

$$\Rightarrow a_n = 3 - n2^{-n} - [3 - (n-1)2^{-(n-1)}]$$

$$= \cancel{3} - n2^{-n} - \cancel{3} + (n-1)2^{-(n-1)}$$

$$= \frac{-n}{2^n} + \frac{n-1}{2^{n-1}}$$

$$= \frac{-n}{2^n} + \frac{2(n-1)}{2^n} = \frac{-n+2n-2}{2^n} = \frac{n-2}{2^n}$$

$$\Rightarrow \text{Sum} = \lim_{n \rightarrow \infty} S_n$$

$$= \lim_{n \rightarrow \infty} (3 - n2^{-n})$$

$$= 3 - 0 = 3$$

Let $f(x) = x2^{-x}$

$$\lim_{x \rightarrow \infty} \frac{x}{2^x} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{1}{2^x \ln 2}$$

$$= \frac{1}{\infty} = 0$$



Text book question



If the n th partial sum of the series $\sum_{n=1}^{\infty} a_n$ is $S_n = \frac{n-1}{n+1}$

Find a_n and $\sum_{n=1}^{\infty} a_n$.

$$a_n = S_n - S_{n-1}$$

$$S_n = \frac{n-1}{n+1} \rightarrow S_{n-1} = \frac{n-1-1}{n-1+1} = \frac{n-2}{n}$$

$$a_n = S_n - S_{n-1}$$

$$= \frac{n-1}{n+1} - \frac{n-2}{n}$$

$$= \frac{n(n-1)}{n(n+1)} - \frac{(n-2)(n+1)}{n(n+1)}$$

$$= \frac{n^2 - n - [n^2 + n - 2n - 2]}{n(n+1)}$$

$$= \frac{\cancel{n^2} - \cancel{n} - \cancel{n^2} - \cancel{n} + 2\cancel{n} + 2}{n(n+1)} = \frac{2}{n(n+1)}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n-1}{n+1} = \lim_{n \rightarrow \infty} \frac{\cancel{n} [1 - \frac{1}{n}]}{\cancel{n} [1 + \frac{1}{n}]} = \frac{1-0}{1+0} = 1$$

$$\therefore \sum_{n=1}^{\infty} a_n = 1$$



Multiple choice



The n th partial sum of the series $\sum_{n=1}^{\infty} \frac{2}{(2n-3)(2n-1)}$ is

a) $-1 + \frac{2}{(2n-3)(2n-1)}$

b) $-1 - \frac{1}{2n-1}$

c) $1 - \frac{1}{2n-3}$

d) $1 + \frac{1}{3} - \frac{1}{2n-3} - \frac{1}{2n-1}$

e) $-1 + 1 - \frac{1}{2n-3} - \frac{1}{2n-1}$

Dec 2016

Let $f(x) = \frac{2}{(2x-3)(2x-1)}$

$$\frac{2}{(2x-3)(2x-1)} = \frac{A}{2x-3} + \frac{B}{2x-1}$$

* $(2x-3)(2x-1)$

$$2 = A(2x-1) + B(2x-3)$$

$$x = \frac{1}{2} \rightarrow 2 = 0 + B(2 \cdot \frac{1}{2} - 3) \rightarrow 2 = -2B \rightarrow B = -1$$

$$x = \frac{3}{2} \rightarrow 2 = A(2 \cdot \frac{3}{2} - 1) + 0 \rightarrow 2 = 2A \rightarrow A = 1$$

$$\therefore a_n = \frac{1}{2n-3} - \frac{1}{2n-1}$$

$$S_n = a_1 + a_2 + \dots + a_{n-1} + a_n$$

$$= \left[\frac{1}{-1} - \frac{1}{1} \right] + \left[\frac{1}{1} - \frac{1}{3} \right] + \dots + \left[\frac{1}{2(n-1)-3} - \frac{1}{2(n-1)-1} \right]$$

$$\therefore S_n = -1 - \frac{1}{2n-1} \quad (b) \checkmark$$



Multiple choice



The series $\sum_{n=1}^{\infty} [\sin n - \sin(n+1)]$

Dec 2017

- a) converges and its sum equal to $\pi/2$.
- b) converges and its sum equal to $\sin 1$.
- c) converges and its sum equal to 0.
- d) diverges.
- e) converges and its sum equal to $-\pi/2$.

$$a_n = \sin n - \sin(n+1)$$

$$S_n = a_1 + a_2 + \dots + a_{n-1} + a_n$$

$$= [\sin 1 - \cancel{\sin 2}] + [\cancel{\sin 2} - \cancel{\sin 3}] + \dots + [\cancel{\sin(n-1)} - \cancel{\sin(n)}] + [\cancel{\sin n} - \sin(n+1)]$$

$$S_n = \sin 1 - \sin(n+1)$$

$$\lim_{n \rightarrow \infty} S_n$$

$$\lim_{n \rightarrow \infty} [\sin 1 - \sin(n+1)]$$

DNE

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is divergent}$$

(Telescoping)



Geometric series



$$\sum (\text{رقم})^n$$

$$\sum 3^n$$

$$\sum 3^{2n+1}$$

$$\sum e^{n-2} 4^n$$

$$\sum \frac{2^n}{5^{2n+3}}$$

خطوات الحل

$$a = a_1$$

1) أعوض في البداية

$$a_2$$

2) أعوض بالرقم الي بعده

$$r = \frac{a_2}{a_1}$$

$$r$$

3) أقسمهم الثاني على الأول وأطلع

if

$$|r| > 1$$

Divergent

NO SUM

$$|r| < 1$$

Convergent

$$S = \frac{a}{1-r}$$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=1}^{\infty} \pi^n e^{-2n-3}$$

$$\underline{n=1} \longrightarrow a_1 = \pi e^{-2-3} = \pi e^{-5} = \frac{\pi}{e^5}$$

$$\implies \therefore a = \frac{\pi}{e^5}$$

$$\underline{n=2} \longrightarrow a_2 = \pi^2 e^{-4-3} = \pi^2 e^{-7} = \frac{\pi^2}{e^7}$$

$$\therefore r = \frac{a_2}{a_1} = \frac{\frac{\pi^2}{e^7}}{\frac{\pi}{e^5}} = \frac{\frac{\pi}{e^2}}{1} = \frac{\pi}{e^2}$$

$$\therefore |r| < 1 \implies \therefore \sum_{n=1}^{\infty} a_n \text{ is convergent}$$

(Geometric Series)

$$\implies S = \frac{a}{1-r} = \frac{\frac{\pi}{e^5}}{1 - \frac{\pi}{e^2}}$$

$$= \frac{\frac{\pi}{e^5}}{\frac{e^2 - \pi}{e^2}} = \frac{\frac{\pi}{e^5}}{e^2 - \pi}$$

$$= \frac{\pi}{e^3 (e^2 - \pi)}$$

Sol #2

$$\sum_{n=1}^{\infty} \pi^n e^{-2n-3}$$

$$\sum_{n=1}^{\infty} \pi^n e^{-2n} e^{-3}$$

$$\sum_{n=1}^{\infty} \pi^n (e^{-2})^n e^{-3}$$

$$\sum_{n=1}^{\infty} \pi^{n-1+1} \left(\frac{1}{e^2}\right)^{n-1+1} e^{-3}$$

$$\sum_{n=1}^{\infty} \pi^{n-1} \pi^1 \left(\frac{1}{e^2}\right)^{n-1} \left(\frac{1}{e^2}\right) \frac{1}{e^3}$$

$$\sum_{n=1}^{\infty} \frac{\pi}{e^2 e^3} \left(\frac{\pi}{e^2}\right)^{n-1} = \sum_{n=1}^{\infty} \frac{\pi}{e^5} \left(\frac{\pi}{e^2}\right)^{n-1}$$

$$a = \frac{\pi}{e^5}, \quad r = \frac{\pi}{e^2}$$

$$|r| = \left| \frac{\pi}{e^2} \right| = \frac{\pi}{e^2} < 1$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent

and the sum $S = \frac{a}{1-r}$

$$S = \frac{\frac{\pi}{e^5}}{1 - \frac{\pi}{e^2}} = \frac{\frac{\pi}{e^5}}{\frac{e^2 - \pi}{e^2}} = \frac{\pi e^2}{(e^2 - \pi) e^5}$$

$$= \frac{\pi}{(e^2 - \pi) e^3}$$

$$\pi \approx 3.14$$

$$e^2 = (2.7)^2$$

Solution #2

$$\sum_{n=1}^{\infty} \pi^n e^{-2n-3}$$

$$a_n = \pi^n e^{-2n-3}$$

$$\begin{aligned} a_{n+1} &= \pi^{n+1} e^{-2(n+1)-3} \\ &= \pi^{n+1} e^{-2n-2-3} \\ &= \pi^{n+1} e^{-2n-5} \end{aligned}$$

$$\begin{aligned} \Rightarrow r &= \frac{a_{n+1}}{a_n} = \frac{\pi^{n+1} e^{-2n-5}}{\pi^n e^{-2n-3}} \\ &= \frac{\cancel{\pi^n} \pi \cancel{e^{-2n}} e^{-5}}{\cancel{\pi^n} \cancel{e^{-2n}} e^{-3}} \\ &= \pi e^{-2} = \frac{\pi}{e^2} \end{aligned}$$

$$\Rightarrow a_1 = \pi e^{-2-3} = \pi e^{-5} = \frac{\pi}{e^5}$$

$$|r| = \left| \frac{\pi}{e^2} \right| = \frac{\pi}{e^2} < 1$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent

and the sum $S = \frac{a}{1-r}$

$$S = \frac{\frac{\pi}{e^5}}{1 - \frac{\pi}{e^2}} = \frac{\frac{\pi}{e^5}}{\frac{e^2 - \pi}{e^2}} = \frac{\pi e^2}{(e^2 - \pi) e^5} = \frac{\pi}{(e^2 - \pi) e^3}$$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$$

$$\underline{n=1} \rightarrow a_1 = 2^2 3^0 = 2^2$$

$$\underline{n=2} \rightarrow a_2 = 2^4 3^{-1} = \frac{2^4}{3}$$

$$\therefore r = \frac{a_2}{a_1} = \frac{\frac{2^4}{3}}{2^2} = \frac{2^2}{3} = \frac{4}{3}$$

$\therefore |r| > 1 \implies \therefore \sum_{n=1}^{\infty} a_n$ is Divergent
(Geometric Series)

$\implies$ No Sum .

AL OTAYBI'S
LIBRARY

Determine whether the series is convergent or divergent .
if converges find its sum

$$\sum_{n=1}^{\infty} (-1)^n \frac{3^{2n}}{e^{n+2}} \quad \rightarrow \quad \sum a_n$$

$$n=1 \longrightarrow a_1 = (-1)^1 \frac{3^2}{e^3} = -\frac{3^2}{e^3}$$

$$n=2 \longrightarrow a_2 = (-1)^2 \frac{3^4}{e^4} = \frac{3^4}{e^4}$$

$$r = \frac{a_2}{a_1} = \frac{\frac{3^4}{e^4}}{-\frac{3^2}{e^3}} = -\frac{3^2}{e} = -\frac{9}{e}$$

$$|r| = \left| -\frac{9}{e} \right| = \frac{9}{e} > 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} (-1)^n \frac{3^{2n}}{e^{n+2}}$ is Divergent
(Geometric Series)

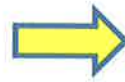
$\Rightarrow$ No Sum.



Series properties



$$\sum [a_n \pm b_n] = \sum a_n \pm \sum b_n$$



يمكن توزيع السلسلة
على الجمع والطرح

$$\sum k a_n = k \sum a_n$$



أقدر أطلع الرقم
المضروب

$$\sum \pm \sum$$

Convergent $\pm$ Convergent



Convergent

Divergent $\pm$ Convergent



Divergent

Divergent $\pm$ Divergent



لا يوجد دلالة !!



Text book question



Determine whether the series is convergent or divergent.

If convergent, find its sum.

$$\sum_{n=1}^{\infty} [(-0.2)^n + (0.6)^{n-1}] \quad \xrightarrow{\text{red arrow}} \quad a_n$$

$$\sum_{n=1}^{\infty} (-0.2)^n \quad + \quad \sum_{n=1}^{\infty} (0.6)^{n-1}$$

$$n=1 \longrightarrow a_1 = -0.2 \quad n=1 \longrightarrow a_1 = 1$$

$$n=2 \longrightarrow a_2 = (-0.2)^2 \quad n=2 \longrightarrow a_2 = (0.6)^1$$

$$r = \frac{a_2}{a_1} = \frac{(-0.2)^2}{-0.2} = -0.2 \quad r = \frac{a_2}{a_1} = \frac{0.6}{1} = 0.6$$

$$\therefore |r| < 1$$

$$\therefore |r| < 1$$

$\therefore \sum_{n=1}^{\infty} (-0.2)^n$ is convergent (Geometric Series) $\therefore \sum_{n=1}^{\infty} (0.6)^{n-1}$ is convergent (Geometric Series)

$$S = \frac{a}{1-r}$$

$$= \frac{-0.2}{1 - (-0.2)}$$

$$= \frac{-\frac{2}{10}}{\frac{10}{10} + \frac{2}{10}}$$

$$= \frac{-\frac{2}{10}}{\frac{12}{10}} = -\frac{1}{6}$$

$$S = \frac{a}{1-r}$$

$$= \frac{1}{1 - 0.6} = \frac{1}{\frac{10}{10} - \frac{6}{10}}$$

$$= \frac{1}{\frac{4}{10}} = \frac{1}{\frac{2}{5}} = \frac{5}{2}$$

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is convergent and Sum} = \frac{-1}{6} + \frac{5}{2} = \frac{-2}{12} + \frac{30}{12} = \frac{28}{12} = \frac{7}{3}$$

Determine whether the series is convergent or divergent .
if converges find its sum

$$\sum_{n=1}^{\infty} \frac{2^n + (-1)^n}{3^n}$$

$\rightarrow \sum a_n$

$$\sum_{n=1}^{\infty} \left[\frac{2^n}{3^n} + \frac{(-1)^n}{3^n} \right]$$

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n + \sum_{n=1}^{\infty} \left(-\frac{1}{3}\right)^n$$

$$n=1 \rightarrow a_1 = \left(\frac{2}{3}\right)^1$$

$$n=1 \rightarrow a_1 = \left(-\frac{1}{3}\right)^1$$

$$n=2 \rightarrow a_2 = \left(\frac{2}{3}\right)^2$$

$$n=2 \rightarrow a_2 = \left(-\frac{1}{3}\right)^2$$

$$r = \frac{a_2}{a_1} = \frac{\left(\frac{2}{3}\right)^2}{\frac{2}{3}} = \frac{2}{3}$$

$$r = \frac{a_2}{a_1} = \frac{\left(-\frac{1}{3}\right)^2}{\left(-\frac{1}{3}\right)} = -\frac{1}{3}$$

$$|r| = \left|\frac{2}{3}\right| < 1$$

$$|r| = \left|-\frac{1}{3}\right| = \frac{1}{3} < 1$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ is convergent
(Geometric Series)

$\therefore \sum_{n=1}^{\infty} \left(-\frac{1}{3}\right)^n$ is convergent
(Geometric Series)

$$S = \frac{a}{1-r} = \frac{\frac{2}{3}}{1-\frac{2}{3}} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2$$

$$S = \frac{a}{1-r} = \frac{-\frac{1}{3}}{1-\left(-\frac{1}{3}\right)} = \frac{-\frac{1}{3}}{\frac{4}{3}} = -\frac{1}{4}$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent

$$S = 2 + \left(-\frac{1}{4}\right) = \frac{8}{4} - \frac{1}{4} = \frac{7}{4}$$

Determine whether the series is convergent or divergent . if converges find its sum

$$\sum_{n=0}^{\infty} \left[\left(\frac{e+1}{e} \right)^{-n} - \frac{1}{(n+2)(n+1)} \right]$$

$$\sum_{n=0}^{\infty} \left(\frac{e+1}{e} \right)^{-n} - \sum_{n=0}^{\infty} \frac{1}{(n+2)(n+1)}$$

$$n=0 \rightarrow a_0 = 1$$

$$n=1 \rightarrow a_1 = \left(\frac{e+1}{e} \right)^{-1} = \frac{e}{e+1}$$

$$r = \frac{a_1}{a_0} = \frac{\frac{e}{e+1}}{1} = \frac{e}{e+1}$$

$$|r| = \left| \frac{e}{e+1} \right| < 1$$

$\Rightarrow \sum_{n=0}^{\infty} a_n$ is convergent (Geometric Series)

$$\rightarrow S = \frac{a}{1-r} = \frac{1}{1 - \frac{e}{e+1}} = \frac{1}{\frac{e+1-e}{e+1}} = e+1$$

$$a_n = \frac{1}{(n+2)(n+1)}$$

$$\frac{1}{(x+2)(x+1)} = \frac{A}{x+2} + \frac{B}{x+1}$$

$$* (x+2)(x+1)$$

$$1 = A(x+1) + B(x+2)$$

$$x = -1 \quad 1 = 0 + B(1) \rightarrow B = 1$$

$$x = -2 \quad 1 = A(-1) + 0 \rightarrow A = -1$$

$$a_n = \frac{-1}{n+2} + \frac{1}{n+1} = \frac{1}{n+1} - \frac{1}{n+2}$$

$$S_n = a_0 + a_1 + \dots + a_{n-1} + a_n$$

$$= \left[\frac{1}{1} - \frac{1}{2} \right] + \left[\frac{1}{2} - \frac{1}{3} \right] + \dots + \left[\frac{1}{n-1+1} - \frac{1}{n-1+2} \right] + \left[\frac{1}{n+1} - \frac{1}{n+2} \right]$$

$$S_n = 1 - \frac{1}{n+2}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+2} \right] = 1 - 0 = 1$$

$\Rightarrow \sum_{n=0}^{\infty} a_n$ is convergent (Telescoping)

$\Rightarrow \sum_{n=0}^{\infty} \left[\left(\frac{e+1}{e} \right)^{-n} - \frac{1}{(n+2)(n+1)} \right]$ is convergent and Sum = $e+1 - 1 = e$



Multiple choice



The sum of the series $\sum_{n=1}^{\infty} 2^{4-3n}$ is

Dec 2016

- a) 16. b) $\frac{128}{7}$ c) $\frac{16}{7}$ d) 0 e) The series diverges.

$$n=1 \longrightarrow a_1 = 2^{4-3} = 2$$

$$n=2 \longrightarrow a_2 = 2^{4-6} = 2^{-2} = \frac{1}{2^2}$$

$$\therefore r = \frac{a_2}{a_1} = \frac{\frac{1}{2^2}}{2} = \frac{\frac{1}{4}}{2} = \frac{1}{8}$$

$|r| < 1 \longrightarrow \therefore \sum_{n=1}^{\infty} 2^{4-3n}$ is convergent
(Geometric Series)

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{2}{1 - \frac{1}{8}} = \frac{2}{\frac{8}{8} - \frac{1}{8}} = \frac{2}{\frac{7}{8}} \\ &= \frac{16}{7} \end{aligned}$$

(c)



Multiple choice



The series $\sum_{n=0}^{\infty} \frac{2^n}{3} 5^{-2n}$

Spring 2019

- a) is divergent. b) has a sum equal to 25/69.
c) has a sum equal to 25/2. d) has a sum equal to 25/6.

$$n=0 \rightarrow a_0 = \frac{2^0}{3} 5^0 = \frac{1}{3}$$

$$n=1 \rightarrow a_1 = \frac{2}{3} 5^{-2} = \frac{2}{3(5^2)}$$

$$\therefore r = \frac{a_1}{a_0} = \frac{\frac{2}{3(5^2)}}{\frac{1}{3}} = \frac{2}{25}$$

$$|r| < 1 \rightarrow \therefore \sum_{n=0}^{\infty} \frac{2^n}{3} 5^{-2n} \text{ is convergent (Geometric Series)}$$

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{\frac{1}{3}}{1 - \frac{2}{25}} = \frac{\frac{1}{3}}{\frac{25}{25} - \frac{2}{25}} = \frac{\frac{1}{3}}{\frac{23}{25}} \\ &= \frac{25}{23(3)} = \frac{25}{69} \quad (b) \end{aligned}$$



The general form of geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots$$

a_1
1st term

a_2
2nd term

a_3
3rd term

a_4
4th term



The first term of geometric series is $a_1 = 24$
and its fourth term is $a_4 = \frac{1}{9}$

Find the sum of the series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + ar^4 + \dots$$

$\downarrow$
 a_1

$\downarrow$
 a_4

$$\Rightarrow a_1 = a = 24$$

$$\Rightarrow a_4 = ar^3 = \frac{1}{9}$$

$$24r^3 = \frac{1}{9}$$

$$r^3 = \frac{1}{9(24)} = \frac{1}{216}$$

$$r^3 = \frac{1}{216} \Rightarrow r = \frac{1}{6}$$

$$S = \frac{a}{1-r}$$

$$= \frac{24}{1 - \frac{1}{6}} = \frac{24}{\frac{6}{6} - \frac{1}{6}} = \frac{24}{\frac{5}{6}} = \frac{144}{5}$$

Find the value of c such that

$$\sum_{n=0}^{\infty} e^{nc} = 10$$

$\sum_{n=0}^{\infty} (e^c)^n$ Geometric Series

$n=0 \longrightarrow a_0 = (e^c)^0 = 1$

$n=1 \longrightarrow a_1 = (e^c)^1 = e^c$

$r = \frac{a_1}{a_0} = \frac{e^c}{1} = e^c$

$S = \frac{a}{1-r} = \frac{1}{1-e^c}$

$$\begin{aligned} \frac{1}{1-e^c} &= 10 \\ 1 &= 10(1-e^c) \\ 1 &= 10 - 10e^c \\ 10e^c &= 9 \\ e^c &= \frac{9}{10} \Rightarrow C = \ln \frac{9}{10} \end{aligned}$$

Find the value of t such that

$$\sum_{n=1}^{\infty} \left(\frac{t}{t+2} \right)^n = 3$$

$\sum_{n=1}^{\infty} \left(\frac{t}{t+2} \right)^n$ Geometric Series

$n=1 \longrightarrow a_1 = \frac{t}{t+2}$

$n=2 \longrightarrow a_2 = \left(\frac{t}{t+2} \right)^2$

$r = \frac{a_2}{a_1} = \frac{\left(\frac{t}{t+2} \right)^2}{\frac{t}{t+2}} = \frac{t}{t+2}$

$S = \frac{a}{1-r} = \frac{\frac{t}{t+2}}{1 - \frac{t}{t+2}} = \frac{\frac{t}{t+2}}{\frac{t+2-t}{t+2}} = \frac{t}{2}$

$\frac{t}{2} = 3 \longrightarrow t = 6$

Find the values of $\theta \in [0, \pi]$ for which the series

$$\sum_{n=1}^{\infty} \cos^n \theta \quad \text{is convergent and find its sum}$$

$$\sum_{n=1}^{\infty} (\cos \theta)^n \quad \text{Geometric Series}$$

$$n=1 \longrightarrow a_1 = (\cos \theta)^1 = \cos \theta$$

$$n=2 \longrightarrow a_2 = (\cos \theta)^2$$

$$r = \frac{a_2}{a_1} = \frac{(\cos \theta)^2}{\cos \theta} = \cos \theta$$

The series is convergent only if

$$|r| < 1$$

$$|\cos \theta| < 1$$

$$-1 < \cos \theta < 1$$

$$\cos^{-1}(-1) < \cos^{-1} \cos \theta < \cos^{-1}(1)$$

$$\pi < \theta < 0 \longrightarrow \theta \in (0, \pi)$$

$$\text{Sum} = S = \frac{a}{1-r} = \frac{\cos \theta}{1-\cos \theta}$$

$$\text{if } \sum_{n=1}^{\infty} \frac{x^n}{10^{2n+1}}$$

(a) Find all values of x for which the series is convergent.

(b) Find the sum of the series.

$$\sum_{n=1}^{\infty} \frac{x^n}{10^{2n+1}} \text{ is geometric series}$$

$$n=1 \longrightarrow a_1 = \frac{x^1}{10^3}$$

$$n=2 \longrightarrow a_2 = \frac{x^2}{10^5}$$

$$r = \frac{a_2}{a_1} = \frac{\frac{x^2}{10^5}}{\frac{x}{10^3}} = \frac{x}{10^2} = \frac{x}{100}$$

The series is convergent only if

$$|r| < 1$$

$$\left| \frac{x}{100} \right| < 1$$

$$-1 < \frac{x}{100} < 1$$

$$-100 < x < 100 \longrightarrow x \in (-100, 100)$$

$$\begin{aligned} \text{Sum} = S &= \frac{a}{1-r} = \frac{\frac{x}{10^3}}{1 - \frac{x}{10^2}} = \frac{\frac{x}{10^3}}{\frac{10^2 - x}{10^2}} \\ &= \frac{\frac{x}{10}}{10^2 - x} = \frac{x}{10(100 - x)} \end{aligned}$$



العدد العشري الغير منتهي !!



Write the number $1.\overline{23} = 1.232323 \dots$
as ratio of integer

$$\begin{aligned}
 1.\overline{23} &= 1 + 0.23 + 0.0023 + 0.000023 + \dots \\
 &= 1 + \frac{23}{10^2} + \frac{23}{10^4} + \frac{23}{10^6} + \dots
 \end{aligned}$$

Geometric Series

$$a = \frac{23}{10^2} \quad r = \frac{\frac{23}{10^4}}{\frac{23}{10^2}} = \frac{1}{10^2}$$

$$\Rightarrow S = \frac{a}{1-r}$$

$$= \frac{\frac{23}{10^2}}{1 - \frac{1}{10^2}} = \frac{\frac{23}{10^2}}{\frac{10^2 - 1}{10^2}} = \frac{23}{100 - 1} = \frac{23}{99}$$

$$\begin{aligned}
 \Rightarrow 1.\overline{23} &= 1 + S \\
 &= 1 + \frac{23}{99} \\
 &= \frac{99}{99} + \frac{23}{99} = \frac{122}{99}
 \end{aligned}$$

Let x be the number having repeated decimal representation $x = 0.794794794 \dots$

(a) Express x as the sum of geometric series

(b) Find the integers m and n such that $x = \frac{m}{n}$

$$x = 0.\underline{794}\underline{794}\underline{794} \dots$$

$$= 0.794 + 0.000794 + 0.000000794 + \dots$$

$$= \frac{794}{10^3} + \frac{794}{10^6} + \frac{794}{10^9} + \dots$$

$$a = \frac{794}{10^3}$$

$$r = \frac{\frac{794}{10^6}}{\frac{794}{10^3}} = \frac{1}{10^3}$$

$$\begin{aligned} (a) \quad x = S &= \frac{a}{1-r} = \frac{\frac{794}{10^3}}{1 - \frac{1}{10^3}} \\ &= \frac{\frac{794}{10^3}}{\frac{10^3}{10^3} - \frac{1}{10^3}} = \frac{\frac{794}{10^3}}{\frac{10^3 - 1}{10^3}} \\ &= \frac{794}{1000 - 1} = \frac{794}{999} \end{aligned}$$

$$(b) \quad x = \frac{794}{999} = \frac{m}{n} \implies \begin{aligned} m &= 794 \\ n &= 999 \end{aligned}$$



Text book question



Determine whether the geometric series is convergent or divergent.

If convergent , find its sum.

$$3 - 4 + \frac{16}{3} - \frac{64}{9} + \dots$$

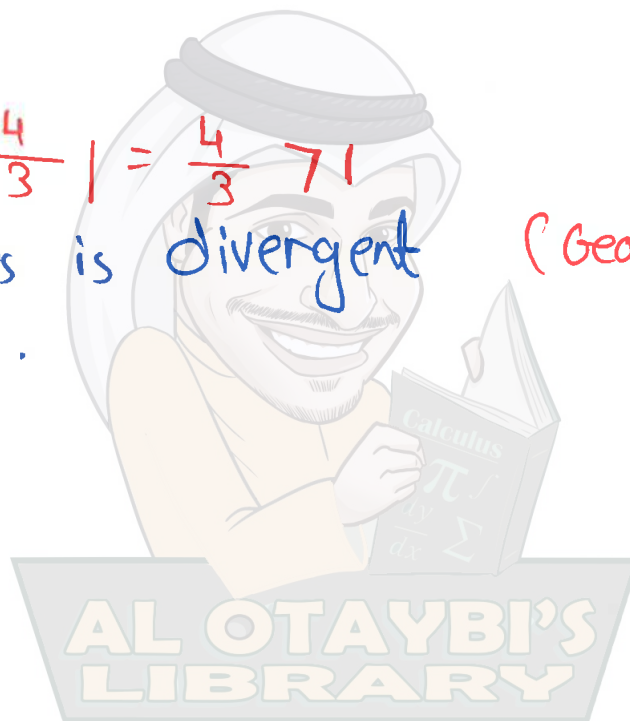
$$a = 3$$

$$r = \frac{-4}{3}$$

$$\therefore |r| = \left| \frac{-4}{3} \right| = \frac{4}{3} > 1$$

$\therefore$ The series is divergent (Geometric Series)

$\therefore$ NO Sum.



Test of Divergence

N^{th} Term Test (N.T.T)

$$\text{if } \sum a_n$$

$$\lim_{n \rightarrow \infty} a_n$$

$$\neq 0$$

$$= 0$$

$$\sum a_n$$

is divergent

فشل الاختبار !!

Examples

Determine whether the series is convergence or divergence

$$\sum_{n=1}^{\infty} \frac{n^2}{5 + 3n^2} \rightarrow \sum a_n$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^2}}{\cancel{n^2} \left[\frac{5}{n^2} + 3 \right]} = \frac{1}{0 + 3} = \frac{1}{3} \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is divergent (N.T.T test of divergence)



Text book question



Determine whether the series is convergent or divergent.
If convergent, find its sum.

$$\sum_{n=1}^{\infty} \arctan n$$

$$\sum_{n=1}^{\infty} \tan^{-1} n \rightarrow a_n$$

$$\lim_{n \rightarrow \infty} \tan^{-1} n = \frac{\pi}{2} \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence) (N.T.T)

Determine whether the series is convergent or divergent.
If convergent, find its sum.

$$\sum_{n=1}^{\infty} n \sin\left(\frac{\pi}{n}\right) \rightarrow a_n$$

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$$

$$\lim_{n \rightarrow \infty} \cancel{n} \cdot \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \cdot \frac{\pi}{\cancel{n}} = (1)\pi = \pi \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence) (N.T.T)

Determine whether the series is convergent or divergent.
If convergent, find its sum.

$$\sum_{n=1}^{\infty} n \tan\left(\frac{1}{\sqrt{n}}\right) \rightarrow a_n$$

$$\lim_{n \rightarrow \infty} n \tan\left(\frac{1}{\sqrt{n}}\right)$$

$$\lim_{n \rightarrow \infty} \cancel{n} \cdot \frac{\tan\left(\frac{1}{\sqrt{n}}\right)}{\frac{1}{\sqrt{n}}} \cdot \frac{1}{\cancel{\sqrt{n}}} = \lim_{n \rightarrow \infty} \sqrt{n} \cdot \frac{\tan\left(\frac{1}{\sqrt{n}}\right)}{\frac{1}{\sqrt{n}}} = \infty (1) = \infty \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence) (N.T.T)

Show that the Series

$$\sum_{n=1}^{\infty} \frac{1}{1 + (0.2)^n} \text{ is divergent}$$

$$\lim_{n \rightarrow \infty} \frac{1}{1 + (0.2)^n} = \frac{1}{1 + 0} = \frac{1}{1} = 1 \neq 0$$

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is divergent.}$$

$$\lim_{n \rightarrow \infty} 0.2^n = 0, |0.2| < 1$$

(Test of divergence) (N.T.T)

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{e^{\sqrt[3]{n}}}{\sqrt[3]{n^2}} \text{ convergent or divergent}$$

$$\lim_{n \rightarrow \infty} \frac{e^{\sqrt[3]{n}}}{\sqrt[3]{n^2}} = \lim_{n \rightarrow \infty} \frac{e^{n^{\frac{1}{3}}}}{n^{\frac{2}{3}}} = \frac{\infty}{\infty}$$

$$\hookrightarrow f(x) = \frac{e^{x^{\frac{1}{3}}}}{x^{\frac{2}{3}}}$$

$$\lim_{x \rightarrow \infty} \frac{e^{x^{\frac{1}{3}}}}{x^{\frac{2}{3}}}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{e^{x^{\frac{1}{3}}} \cdot \frac{1}{3} x^{-\frac{2}{3}}}{\frac{2}{3} x^{-\frac{1}{3}}} = \lim_{x \rightarrow \infty} \frac{e^{x^{\frac{1}{3}}} x^{-\frac{1}{3}}}{2}$$

$$\lim_{x \rightarrow \infty} \frac{e^{x^{\frac{1}{3}}}}{2 x^{\frac{1}{3}}} = \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{e^{x^{\frac{1}{3}}} \cdot \frac{1}{3} x^{-\frac{2}{3}}}{2 \cdot \frac{1}{3} x^{-\frac{2}{3}}} = \frac{\infty}{2} = \infty \neq 0$$

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is divergent (N.T.T) (Test of divergence)}$$

Determine whether the series is convergence or divergence

$$\sum_{n=1}^{\infty} \left(\frac{n+3}{n-2} \right)^{n+2} \longrightarrow \sum_{n=1}^{\infty} a_n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{n+3}{n-2} \right)^{n+2} &= \lim_{n \rightarrow \infty} \left(\frac{n-2+5}{n-2} \right)^{n+2} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n-2+5}{n-2} \right)^{n+2} = \lim_{n \rightarrow \infty} \left(\frac{n-2}{n-2} + \frac{5}{n-2} \right)^{n+2} \\ &= \lim_{n \rightarrow \infty} \left(1 + \frac{5}{n-2} \right)^{n+2} = e^{\frac{5}{1}} = e^5 \neq 0 \\ \implies \therefore \sum_{n=1}^{\infty} a_n \text{ is divergent (N.T.T test of divergence)} \end{aligned}$$

Determine whether the series is convergence or divergence

$$\sum_{n=1}^{\infty} 5^n \tan(2^{-n}) \longrightarrow \sum_{n=1}^{\infty} a_n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} 5^n \tan(2^{-n}) &\quad \begin{matrix} \infty \cdot \tan 0 \\ \infty \cdot 0 \end{matrix} \\ &= \lim_{n \rightarrow \infty} \frac{5^n \tan(2^{-n})}{2^{-n}} \cdot 2^{-n} \\ &= \lim_{n \rightarrow \infty} 5^n 2^{-n} \frac{\tan(2^{-n})}{2^{-n}} = \lim_{n \rightarrow \infty} \frac{5^n}{2^n} \frac{\tan(2^{-n})}{2^{-n}} \\ &= \lim_{n \rightarrow \infty} \left(\frac{5}{2} \right)^n \frac{\tan(2^{-n})}{2^{-n}} = \infty (1) = \infty \neq 0 \\ \implies \therefore \sum_{n=1}^{\infty} a_n \text{ is divergence (N.T.T test of div.)} \end{aligned}$$

Determine whether the series

$$\sum_{n=0}^{\infty} \frac{n - \pi^n}{n^2 + e^n} \quad \text{convergent or divergent}$$

$$\lim_{n \rightarrow \infty} \frac{n - \pi^n}{n^2 + e^n}$$

$$\div e^n$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n}{e^n} - \frac{\pi^n}{e^n}}{\frac{n^2 + e^n}{e^n}}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n}{e^n} - \left(\frac{\pi}{e}\right)^n}{\frac{n^2}{e^n} + 1}$$

$$\lim_{n \rightarrow \infty} \frac{n}{e^n}$$

$$\hookrightarrow f(x) = \frac{x}{e^x}$$

$$\lim_{x \rightarrow \infty} \frac{x}{e^x}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\infty} = 0$$

$$\lim_{x \rightarrow \infty} \frac{n^2}{e^n}$$

$$\hookrightarrow f(x) = \frac{x^2}{e^x}$$

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow \infty} \frac{2x}{e^x} = \frac{\infty}{\infty}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow \infty} \frac{2}{e^x} = \frac{2}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \left(\frac{\pi}{e}\right)^n$$

$$= \infty$$

$$|\pi/e| > 1$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\frac{n}{e^n} - \left(\frac{\pi}{e}\right)^n}{\frac{n^2}{e^n} + 1} = \frac{0 - \infty}{0 + 1} = -\infty \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (N.T.T)
Test of divergence

Determine whether the series is convergent or divergent.

If convergent, find its sum.

$$\sum_{n=1}^{\infty} \left(\cos \left(\frac{1}{n} \right) \right)^n \rightarrow a_n$$

Past exam

$$\lim_{n \rightarrow \infty} \left(\cos \frac{1}{n} \right)^n = \left(\cos \frac{1}{\infty} \right)^{\infty} = (\cos 0)^{\infty} = 1^{\infty}$$

Let $f(x) = \left(\cos \frac{1}{x} \right)^x$

$$y = \left(\cos \frac{1}{x} \right)^x$$

$$\ln y = \ln \left(\cos \frac{1}{x} \right)^x$$

$$\ln y = x \ln \left(\cos \frac{1}{x} \right)$$

$$\lim_{x \rightarrow \infty} x \ln \left(\cos \frac{1}{x} \right) \rightarrow \infty \cdot 0$$

$$\lim_{x \rightarrow \infty} \frac{\ln \left(\cos \frac{1}{x} \right)}{\frac{1}{x}} \rightarrow \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow \infty}$$

$$\frac{-\sin \frac{1}{x} \cdot \left(-\frac{1}{x^2} \right)}{\cos \frac{1}{x} \cdot \left(-\frac{1}{x^2} \right)}$$

$$\lim_{x \rightarrow \infty} -\tan \frac{1}{x} = -\tan 0 = 0$$

$$\therefore \lim_{n \rightarrow \infty} \left(\cos \frac{1}{n} \right)^n = e^0 = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent

(Test of divergence)
N.T.T

Multiple choice

The series $\sum_{n=0}^{\infty} e^{\frac{3}{n^2+1}}$

Spring 2019

a) has a sum equal to 0.

b) converges.

c) diverges

d) has a sum equal to 1.

$$\lim_{n \rightarrow \infty} e^{\frac{3}{n^2+1}} = e^{\frac{3}{\infty}} = e^0 = 1 \neq 0$$

$\therefore \sum_{n=0}^{\infty} a_n$ is divergent (Test of divergence)



اختبارات سريعة



Harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ is divergent}$$

P-series series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

if

$$p > 1$$

Convergent

$$p < 1$$

Divergent

Example

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

Divergent
(Harmonic)

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

$P = 3 > 1$
 $\therefore \sum_{n=1}^{\infty} \frac{1}{n^3}$ is Convergent
 (P-series)

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ $P = \frac{1}{2} < 1$
 $\therefore \sum_{n=1}^{\infty} a_n$ is Divergent
 (P-series)



Text book question



Determine whether the series is convergent or divergent.

If convergent, find its sum.

$$\sum_{n=1}^{\infty} \left(\frac{3}{5^n} + \frac{2}{n} \right) \quad a_n$$

$$\sum_{n=1}^{\infty} \frac{3}{5^n} + \sum_{n=1}^{\infty} \frac{2}{n}$$

$$n=1 \longrightarrow a_1 = \frac{3}{5}$$

$$n=2 \longrightarrow a_2 = \frac{3}{5^2}$$

$$r = \frac{a_2}{a_1} = \frac{\frac{3}{5^2}}{\frac{3}{5}} = \frac{1}{5}$$

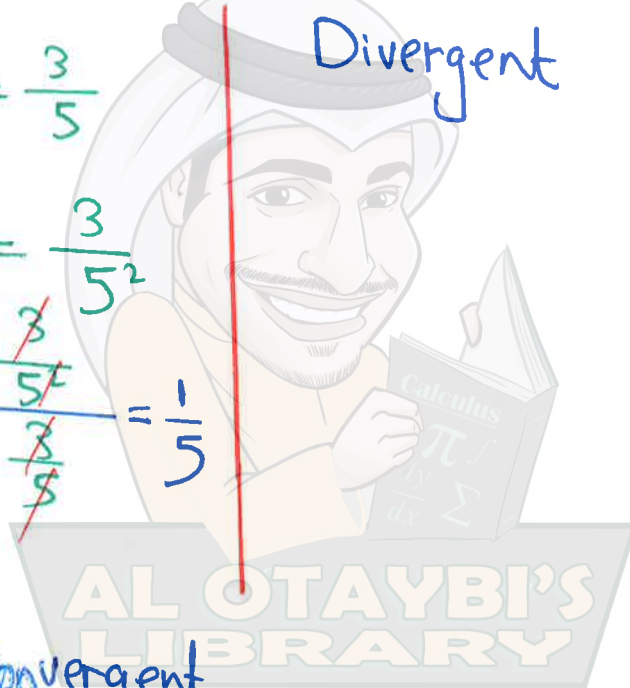
$$\therefore |r| < 1$$

$$\therefore \sum_{n=1}^{\infty} \frac{3}{5^n} \text{ is convergent}$$

(Geometric Series)

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is divergent}$$

$\therefore$ No Sum.





True or False



if $\sum_{n=1}^{\infty} a_n$ is convergent

then $\sum_{n=1}^{\infty} \cos(a_n)$ is also convergent.

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent $\longrightarrow \therefore \lim_{n \rightarrow \infty} a_n = 0$

Now

$$\sum_{n=1}^{\infty} \cos(a_n)$$

$$\lim_{n \rightarrow \infty} \cos(a_n) = \cos(0) = 1 \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \cos(a_n)$ is divergent (N.F.T test of div.)

$\Rightarrow \therefore$ False



Theorem



if the series $\sum_{n=1}^{\infty} a_n$ is convergent

Then



$$\lim_{n \rightarrow \infty} a_n = 0$$

العكس غير صحيح



Example



if $\sum_{n=1}^{\infty} a_n$ is convergent

Determine the limit of the sequence b_n if it exist where

$$b_n = \frac{2 + 3a_n}{1 + a_n^3}$$

$$\because \sum_{n=1}^{\infty} a_n \text{ is convergent} \implies \because \lim_{n \rightarrow \infty} a_n = 0$$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{2 + 3a_n}{1 + a_n^3} = \frac{2 + 3(0)}{1 + 0} = \frac{2 + 0}{1} = 2$$